A Better U VR Companion — Meta Hackathon Application

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Summary

A Better U VR puts a Muse Spark-powered AI coach inside a Meta Quest headset: it watches your reps, talks you through the set in real time by voice, and syncs every rep back into your A Better U account the moment you finish. It's the entry point into a bigger vision — an immersive personal trainer today, a fully linked virtual gym with live remote coaches tomorrow — but the hackathon itself is about proving that one coach works.

Problem & vision

Existing VR fitness apps are closed loops: your workout data stays inside that one app, disconnected from your nutrition, your sleep, your program history, everywhere else you track your health. A Better U already solves that outside the headset, with an exercise library of 1,000 movements, an AI program builder, workout logging and a plate calculator. Putting a coached VR layer on top of that, instead of beside it, is the opening existing competitors have left.

The long-term vision has two parts:

1. **Solo immersive training** — leg-day, body-fit style workouts in VR, fully linked to a person's A Better U account and data rather than a standalone fitness app.
2. **Virtual coached gym** — a coach anywhere in the world (London, say) runs a live virtual session for people joining from anywhere else (Australia, say), all inside a shared virtual gym space, all logged back to A Better U.

The hackathon is the forcing function to build the first part now and prove the concept works before committing to the much bigger multiplayer build.

What we're building for the hackathon

Meta AI's own suggested scope for this track was a simpler voiceover-guided workout with logging and a dashboard. We're pitching the full two-part vision above, but scoping the 10-day build to one demoable slice: a single bodyweight exercise loop (e.g. squats) in a

solo VR scene, with a Muse Spark-powered coach that:

- Talks the user into position and starts the set by voice
- Listens continuously (full voice-to-voice: mic in, Muse Spark reasoning, spoken coaching out — no push-to-talk, no on-screen-only captions)
- Comments on pace, form cues and rep count as the set happens, the way a real trainer would
- Logs the completed set back to the user's A Better U account

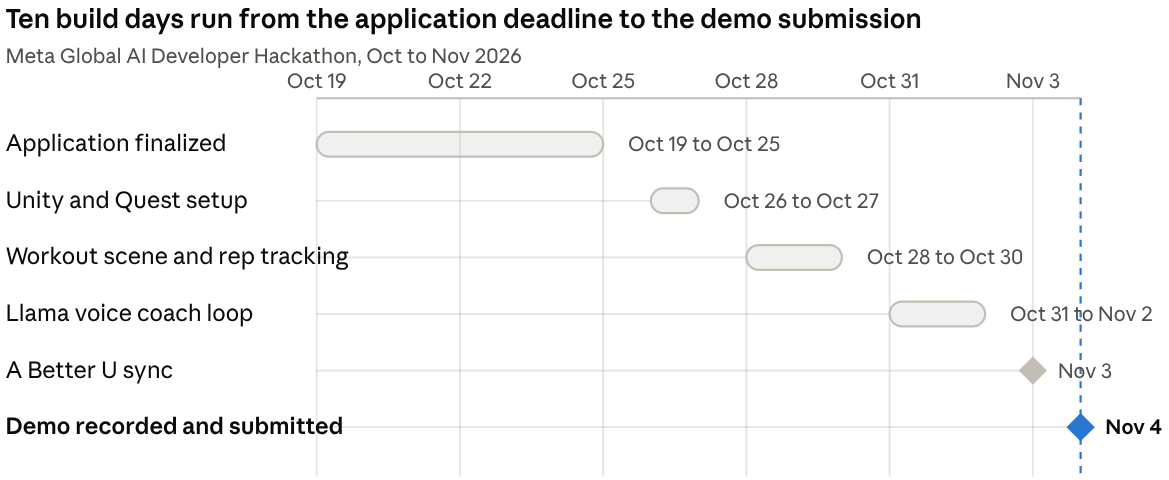
The live multiplayer coached gym is scoped out of the 10-day build entirely and presented in the application as the roadmap beyond the hackathon, not something we claim to demo.

Technical approach

Layer	Choice	Notes
Engine	Unity	Standard for Quest development; new to this team, so the 10-day plan budgets ramp-up time
Hardware	Meta Quest 3S	Owned and available for testing throughout the build
Developer access	Meta developer account	Already set up
AI coach	Muse Spark via Meta's Model API	$150 in Model API credits included with hackathon participation
Voice in	On-device speech-to-text	Captures the user's speech continuously during the set
Voice out	Text-to-speech	Spoken coaching lines, not on-screen captions
Data	A Better U API	Completed sets write back to the user's existing account rather than a siloed VR-only log

Voice loop: mic capture → speech-to-text → prompt assembled with rep count/pace context → Muse Spark call → response → text-to-speech → spoken in-headset. This loop is the core technical risk of the build (latency has to stay low enough that coaching feels live, not delayed), so it's the first thing prototyped, before polishing the workout scene itself.

Claude Code is doing the main implementation work, the same way it's building the rest of A Better U, with this being its first VR/Unity project.



build timeline · 6 milestones, Oct 19 – Nov 4

Applications close Oct 25; the build window itself is Oct 26 – Nov 4. The voice loop (speech-to-text → Muse Spark → text-to-speech) is prototyped early, inside the setup phase, since it's the main technical risk.

Theme relevance

Judging criterion	How this project addresses it
Project clarity	One sentence pitch, one demoable slice (solo squat workout + voice coach), one clear roadmap beyond it (multiplayer coached gym)
Feasibility	Scope is deliberately narrowed from the full vision to a single exercise loop; voice pipeline risk is prototyped first, not last

Judging criterion	How this project addresses it
Theme relevance	Muse Spark is the coach, not a bolt-on feature — it drives the entire in-headset interaction, using Meta's own Model API on Meta's own hardware (Quest 3S)
Technical skill	Real-time voice-to-voice AI inside a VR scene, synced to an existing production platform (A Better U) rather than a standalone demo with no data behind it

The application pitches the full two-part vision (solo training now, live coached multiplayer gym later) to show the bigger opportunity, while being explicit that only the solo slice is built in the 10 days — judges should expect an honest, working demo, not an overclaimed one.

Team & background

Built by the founder of A Better U, an existing fitness platform already covering a 1,000-exercise library, an AI program builder, workout logging, a food database, hormone and sleep tracking, and more — so this isn't a from-scratch fitness app, it's a new front end onto a platform that already has the data and the users' habits behind it. This is the first VR/Unity project for the team; Claude Code is doing the core implementation work, as it does across the rest of the A Better U platform.